

REVIEW

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Focus on the Raman 2D band of the graphene materials: bibliographic review and guiding tools for determining number of layers and stacking configurations

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E-mail: romain.parret@univ-amu.fr and cedric.pardanaud@univ-amu.fr**Keywords:** Raman spectroscopy, graphene, 2D band, twisted grapheneSupplementary material for this article is available [online](#)

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Abstract

Raman spectroscopy is a widely used technique for the analysis of graphene materials within single academic and industry contexts. This technique is characterized by its ease of implementation, rapidity, and non-destructive nature. Spectra resulting from this technique typically consist of two bands (G and 2D), which gives the spectra a seemingly simple appearance. Indeed, as early as 2007, Raman criteria were proposed to determine the number of layers in a stack based solely on the Raman spectrum. However, a multitude of studies published since 2007 have demonstrated that behind this apparent simplicity lie multiple effects that affect the G and 2D bands, thereby rendering interpretation complex and the determination of the number of layers in a stack uncertain. Furthermore, Raman spectroscopy has emerged as a pivotal technique for the analysis of twisted structures and diverse stacking sequences, such as ABA and ABC, which have culminated in significant discoveries, including strongly correlated states such as unconventional superconductivity. In addition to the resonance effects associated with superlattice formation, the shape of the 2D band provides valuable insight into stacking types, although its interpretation remains complex. In this article, we propose a methodology for interpreting the 2D Raman band that is informed by a review of selected references of the extant literature as well as original data. A compendium of recommendations and a series of diagrams are also provided to address other physical effects that can complicate spectral interpretation.

1. Introduction

The discovery of strongly correlated states in twisted bilayers graphene (tBLG) has opened the door to remarkable phenomena, including the prospect of high-temperature superconductivity and macroscopic coherent quantum phases [1–4]. Beyond these unconventional phenomena, the twist angle between adjacent graphene layers has emerged as an effective tuning parameter that governs interlayer interaction and strongly reconstructs the electronic band structure. In particular, the Fermi velocity, which is 10^6 m s^{-1} in single layer graphene (SLG), is continuously renormalized and ultimately suppressed at the so-called magic angle of $\sim 1.1^\circ$, giving rise to flat electronic bands [5]. Recent study [6] also show that within a broad twist-angle range of about $9\text{--}30^\circ$, tBLG (exhibits markedly enhanced electrical transport properties such as higher conductivity and mobility compared with SLG. A further hallmark is the formation of van Hove singularities in the density of states, whose energetic position can be widely tuned, thereby enabling angle-dependent optical resonances spanning an unusually broad spectral range [7–9].

Currently, advanced dry-transfer techniques typically based on polymer stamps enables the fabrication of high-quality twisted graphene heterostructures from exfoliated flakes, with twist angles that can be controlled to within a fraction of a degree [10]. However, the realization of reproducible and well-defined devices critically depends on the prior identification of SLG regions with minimal defects and contamination, as well as on post-assembly characterization of the resulting stacking order and interlayer alignment.

In this context, Raman spectroscopy is one of the most powerful techniques for probing sp^2 -bonded carbon systems. It provides a highly sensitive means of determining both the number of graphene layers and their stacking configuration [10–19]. In its simplest single-wavelength configuration, which is the most widely used in laboratories, it typically reveals essentially the principal features, namely the G and 2D bands. Although seemingly straightforward, interpreting the Raman spectrum is particularly challenging due to the pronounced sensitivity of these two bands to a wide range of factors, including strain [20–22], doping [23–27], the dielectric environment [28], adsorbates [29, 30], structural defect [31–34], edges [35] resonance effects [36–38], non adiabatic effects [39, 40], sample heterogeneity [41] and substrate interactions [42–44].

Here, we focus on the 2D band, a sensitive probe of both layer number and stacking arrangement, making it especially informative for twisted graphene structures. Its lineshape reflects interlayer interactions: multi-component in AB-stacked layers [11, 12], distinct in ABC stacking [45, 46] and Lorentzian in twisted configurations, where the weakened interlayer coupling can make it deceptively similar to that of a SLG [14, 47, 48]. In practice, experimentalists studying graphene and using Raman spectroscopy as a characterization tool often encounter significant challenges in interpreting their spectra, particularly if they are not specialists. We address here these difficulties, which stem from a vast and widely dispersed literature. The interpretation of the 2D band, resulting from a double resonance phenomenon, took several decades of research and numerous debates before reaching a widely accepted consensus [11, 37, 49]. This evolution has been largely driven by the discovery of numerous graphitic structures over time, from the first graphite syntheses and carbon nanotubes to SLG and more recently, tBLG. Another challenge stems from the widespread use, over the past decade, of Raman-based criteria for determining the number of graphene layers. Certain metrics, such as the relative intensity ratio of the 2D to G band (I_{2D}/I_G) or the frequency plot of the 2D to G band (ω_{2D} vs ω_G), have become widely disseminated across the many scientific domains in which graphene is studied. It has become difficult to clearly communicate the limitations of these criteria. It is now well established that a thorough characterization of graphene requires consideration of all spectral features of the 2D band intensity, frequency, linewidth, and lineshape. In this report, we guide the reader through the key articles in the literature related to the 2D band and introduce updated graphical tools to enable more accurate and reliable characterization of graphene materials.

2. Literature review

This section aims to provide a comprehensive overview of the existing literature on graphene materials, focusing on potential challenges that may arise during the research process. Key challenges include the sheer volume of publications, which can obscure important information, variations in specialized terminology across different scientific fields, the constant influx of new findings and the occasional neglect of older studies, which often contain valuable insights that remain relevant today. These difficulties become even more pronounced when considering the Raman 2D band, for which we present the history of the model's development. The complexity of this topic further emphasizes the need for a careful and structured review of the available sources.

2.1. Huge number of seminal important scientific papers

Raman spectroscopy has emerged as a central tool in graphene research. A search for 'Graphene' in ISI-WOS returns 375 000 papers, of which 38 000 also mention 'Raman' (figure 1), a proportion nearly three times higher than that for silicon, where only 25 000 of 684 000 papers include Raman. This highlights that Raman is far more than a routine characterization technique for graphene, it provides a window into the material's fundamental properties. However, the sheer volume of publications presents a challenge when searching for specific information. Limiting searches to papers with both 'Raman' in the title and the material name reduces the set to roughly 1700, and including only reviews makes it manageable for newcomers, though reviews cannot capture the full breadth of the field. Bibliometric analysis further helps identify seminal works, such as A. C. Ferrari's highly cited paper ($\sim 14\,000$ citations) [11], which remains foundational for studies of mono- and few-layer graphene (FLG). Examining recent literature (post-2020), 400 of approximately 1000 papers explicitly mention both Raman and graphene in their

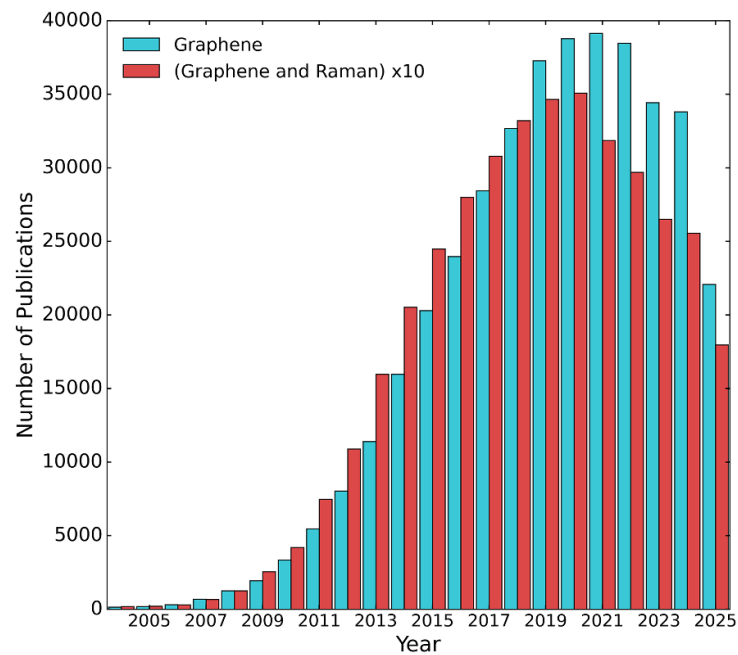


Figure 1. Annual number of publications on graphene and Raman (red) compared with total graphene publications (teal), based on Web of Science, data gathered on 21 September 2025.

titles, indicating that nearly 40% of current studies continue to rely on the technique. These findings highlight both the indispensable role of Raman spectroscopy and the ongoing challenge of navigating the rapidly expanding body of knowledge.

2.2. Specific vocabulary

The difficulty of accessing relevant papers in this field is not only due to the high publication rate discussed earlier. Another contributing factor is the heterogeneity of vocabulary (see a comprehensive analysis in [50]), particularly regarding carbon materials. The term ‘graphene’ is generally used to refer to a SLG, while structures with multiple layers are classified as FLG (2–5 layers), multi-layer graphene (6–10 layers), or graphite (> 10 layers), depending on the number of layers. Establishing this terminology took years, and it is still not universally adopted. Before this consensus emerged, various terms were in use, many of which remain in bibliographic databases. For example, a 2007 paper on bilayer graphene that included Raman studies [51], did not use the term ‘graphene’ in its title but instead referred to ‘graphite,’ as the terms bilayer graphite and bilayer graphene were used interchangeably.

In 2007, 50% of such references used ‘bilayer graphite,’ dropping to 10% in 2014 and 6% in 2021. Consequently, newcomers may overlook key foundational papers simply because terminology has evolved. Adding to this complexity, the use of different graphene synthesis methods, particularly those involving chemical modifications, introduces additional variations in terminology [52]. Materials such as graphene oxide and reduced graphene oxide, while closely related to graphene, fall into categories that will not be discussed here.

2.3. Scientific and technical evolution generating trendy subjects

As science evolves, new concepts and terminology emerge over time. The idea of a twisted graphene bilayer was first introduced in 2007 [53] shortly after the isolation of SLG. A few years later, in 2011, the origin of Moiré patterns and the existence of ‘magic angles,’ where the Fermi velocity vanishes, were established, revealing remarkable properties [54, 55]. Yet, a bibliographic search for titles containing both ‘Twisted graphene’ and ‘Raman’ returns only approximately 100 papers, with the most cited referenced ~400 times [56]. Such a citation number is modest compared to the 50 000 citations of the first 35 papers cited earlier, suggesting that a significant portion of the Raman spectroscopy of graphene community has not yet fully recognized the connection between twist and graphene Raman studies.

These initial subsections demonstrate that extracting relevant information based on bibliometric analysis is challenging. As we will see, this difficulty is even greater when seeking information specifically about the 2D band, since understanding it requires considering studies predating the discovery of

graphene, on other graphene-related structures such as graphite, disordered graphite, or carbon nanotubes. Moreover, the designation of this band has evolved over time: it was referred to as D' or D^* during the 1980s and 1990s, before the term 2D became widely adopted around 2007.

2.4. Connecting old and recent research

At first glance, one might think that the study of stacking faults in carbon is a recent topic, emerging with research on tBLG in the 2010s. In reality, this subject dates back to the late 1970s, when Raman spectroscopy was already being used to study carbon structures. A few years after the seminal paper by Tuinstra *et al* [57] which presented the first Raman spectra of aromatic carbon materials, Oberlin and collaborators investigated carbonization and graphitization, examining the structural evolution of carbons under thermal treatment. Using transmission electron microscopy and Raman spectroscopy, they showed how small graphenic entities (< 1 nm) progressively organize into stacked structures, leading first to turbostratic graphite and, at higher temperatures, to AB-type stacking [58, 59]. In these early studies, the 2D band itself was not investigated.

By the 1980s, several works reported the three characteristic Raman bands of graphitic structures, the D, G, and 2D bands [60, 61]. In particular, the 2D band was observed to exhibit a Lorentzian profile at the early stages of graphitization and to evolve into a bimodal shape as AB stacking developed. Twenty years later, Burian *et al* studied the graphitization of anthracene and saccharose. For saccharose processed at 2300 °C, they observed a Lorentzian 2D band [62]. These pioneering studies already demonstrated the relationship between the shape and width of the 2D band and the degree of disorder in graphitic structures, a result later confirmed, decades afterward, by Cançado *et al* [63].

Despite their importance, these findings are rarely acknowledged in modern discussions of interlayer interactions in graphene. The following paragraph presents the double-resonance mechanism underlying the 2D band. The formulation of this mechanism, and its acceptance by the Raman spectroscopy community working on carbon-based structures, is also the result of a long history over several decades, with current studies on twisted graphene providing new confirmations.

2.5. Origin of 2D band

In the early literature, and often thereafter, the 2D band was described simply as the overtone of the D band. However, this description alone is incomplete. In 1982, Al Jishy and Dresselhaus proposed a phonon model that already attributed the 2D band to zone-boundary phonons at K point [64]. Within this framework, the 2D band shape was explained by the phonon density of states, as is generally the case for second-order processes. This interpretation was taken up again in 1998 by Tan *et al*, who, by measuring the Raman spectrum of highly oriented pyrolytic graphite with different laser lines, discovered an anomalous Raman effect on the 2D band: a frequency shift between the Stokes and anti-Stokes components, together with a dependence on excitation wavelength [65]. This example, now understood within the framework of the double-resonance model, illustrates how early conclusions had to be reconsidered. Indeed, the true origin lies in a double-resonance scattering mechanism, first proposed by Thomsen and Reich and later refined by many groups [11, 37, 38].

In this process, an incoming photon excites an electron-hole pair near the K point of the Brillouin zone. The excited carrier is scattered by a phonon with high momentum $\vec{q}$ (see figure 2(a)), instead of requiring a defect to relax momentum conservation (as in the case of the D band), a second scattering event with a phonon of opposite momentum restores resonance. The electron-hole pair then recombines, emitting a Raman-shifted photon (figure 2(a)). Because two phonons are involved, momentum conservation is satisfied without the need for disorder. Since this mechanism occurs close to the K point, the process remains resonant, but the specific phonons that satisfy the energy- and momentum-conservation rules depend on the excitation energy. Given that the phonon dispersion near the K point displays a Kohn anomaly [66], the 2D band exhibits a pronounced dispersion with the laser excitation energy. This dispersion is approximately $100 \text{ cm}^{-1} \text{ eV}^{-1}$ in the visible range. In AB-stacked bilayer graphene, the splitting of the bands gives rise to four possible double-resonance paths, involving phonons with wavevectors $\vec{q}_{1A}$, $\vec{q}_{1B}$, $\vec{q}_{2A}$, and $\vec{q}_{2B}$ (green arrows in figure 2(b)). This mechanism accounts for the multimodal profile of the 2D band in AB-BLG, which is therefore composed of four components [11, 12]. In tBLG, the reduction of the Fermi velocity modifies the electronic band structure, which in turn changes the double-resonance conditions. For a given laser wavelength, this reduction requires phonons with larger wavevectors to satisfy momentum conservation, compared to monolayer graphene. Therefore, the 2D band experiences a blue shift, reflecting the involvement of higher-energy phonons in the scattering process (figure 2(a)) [12, 19, 48].

This makes the 2D band a highly sensitive probe of the electronic and vibrational properties of graphene [11, 12, 67, 68]. The 2D band has become a key tool for characterizing graphene-based

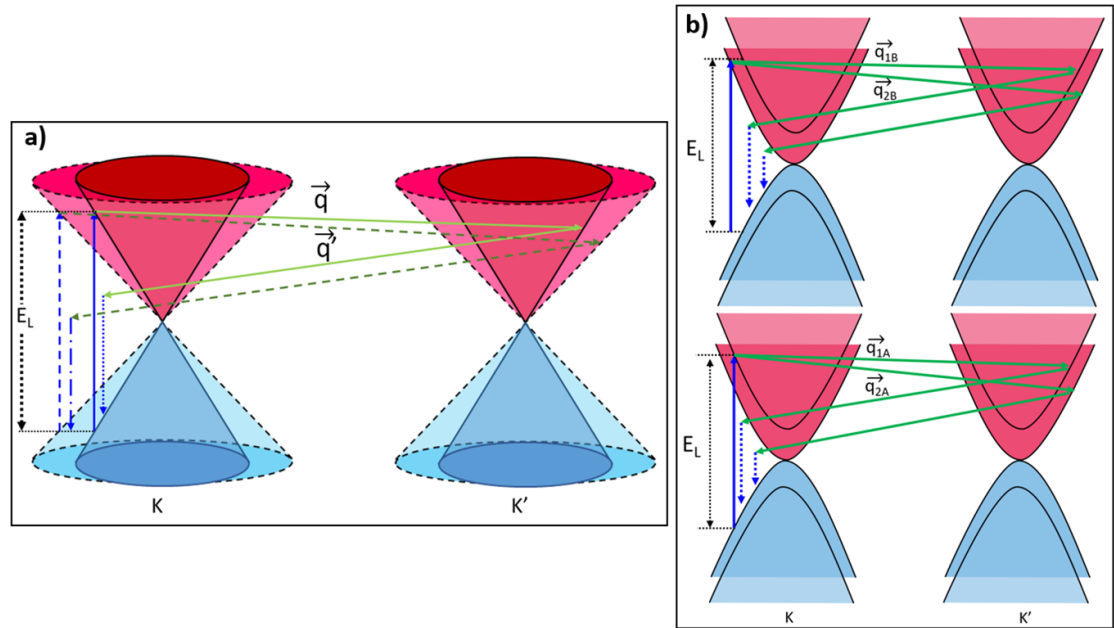


Figure 2. Double-resonance processes in graphene systems. (a) Band structure SLG at the K and K' points. Solid arrows show double-resonance processes involving two phonons with wavevector $\vec{q}$, while dashed arrows illustrate the modified processes ($\vec{q}'$) in tBLG, where the Fermi velocity is reduced due to the smaller band slope near the Dirac points. (b) Band structure of AB-BLG. Green arrows represent double-resonance scattering involving four possible phonons with wavevectors $\vec{q}_{1A}$, $\vec{q}_{2A}$, $\vec{q}_{1B}$, and $\vec{q}_{2B}$.

samples, providing insights into the number of layers and their stacking configuration. However, the 2D band is also sensitive to several factors, such as strain and doping, which complicates its interpretation. In the following subsection, we discuss the use of the I_{2D}/I_G intensity ratio and its reliability for identifying SLG, before addressing the question of whether a Lorentzian 2D profile is uniquely characteristic of SLG.

3. Spectroscopic parameters

3.1. Commonly used 2D band parameter I_{2D}/I_G for identifying a SLG

Ferrari's seminal 2006 paper first reported the Raman spectrum of SLG on Si/SiO₂ substrate [11]. A natural question followed: does graphene exhibit the same behavior when suspended as when supported? The answer, demonstrated in 2009 was no [43]. Suspended graphene was found to be essentially undoped, whereas graphene on SiO₂ exhibit doping that modified all Raman spectroscopic parameter. In particular, the 2D band, which shows a Lorentzian shape in supported graphene, was found to be intrinsically non-Lorentzian in suspended graphene [44]. The 2D band has been shown to be composite, arising from inner and outer scattering processes within the double-resonance mechanism and involving different regions of Dirac cones [44, 69, 70]. Doping affects not only the positions of the G and 2D band but also their relative intensity ratio, I_{2D}/I_G . A commonly used benchmark for identifying high-quality SLG is an I_{2D}/I_G ratio of about 3. In practice, this value can vary at intermediate doping levels [71, 72], and deviations are particularly pronounced in CVD-grown graphene [73]. Furthermore, experimental factors such as polarization mixing at the air-dielectric interface [74], or chromatic aberration of the optical setup [75] can also influence the measured ratio. Several studies have reported I_{2D}/I_G ratios significantly higher than the commonly cited value of ≈ 3 , with strong dependence on substrate and measurement conditions. For example, ratios of ≈ 4 – 5 have been observed on non-flat substrates [76], while an intrinsic value of ≈ 5 has been reported for suspended graphene [77]. The thickness of the SiO₂ layer can also enhance the Raman response and change this ratio [78]. Remarkably, values as high as $I_{2D}/I_G \approx 17$ have been measured in twisted multilayer graphene (MLG), where the 2D band still retains a Lorentzian profile [79]. These results highlight that a Lorentzian 2D profile and an I_{2D}/I_G ratio near 3 are not reliable criteria for identifying SLG.

3.2. 2D band shape vs number of layers

In AB-stacked bilayer, trilayer, and four-layer graphene, the 2D band no longer exhibits a simple Lorentzian profile [12]. Instead, interlayer interactions split the π and π^* electronic bands, giving rise to

multiple components through the double-resonance mechanism. These early works established a foundation for understanding the 2D band in AB-stacked graphene but could not account for spectra when the stacking arrangement is altered. In late 2006, Gupta *et al* linked the 2D band to the number of graphene layers, reporting a Lorentzian shape for both SLG and BLG, with additional components appearing only from three layers onward [80]. In 2007, Graf *et al* observed variations in the 2D profile, presenting a composition of four components in some cases but not in others [41]. A year later, Poncharal *et al* reported Lorentzian 2D band for bilayer graphene [47]. This led to a comment by Ni's group and a reply [81, 82], which we discuss in the next section.

A more systematic approach was introduced in 2010 by Hao *et al* [15] who performed a statistical analysis of mechanically exfoliated flakes. They demonstrated that the number of layers strongly influences the 2D band shape: the bandwidth ranged between 22 and 30 cm^{-1} for SLG, increasing to 50, 57, and 62 cm^{-1} for bilayers, trilayers, and quadrilayers, respectively, with clearly composite profiles. Interestingly, some few-layer flakes still exhibited nearly Lorentzian 2D bands, suggesting a more complex underlying mechanism, this observation is largely overlooked at the time. They also confirmed that the 2D/G intensity ratio decreases systematically with increasing layer number.

In 2015, Bayle *et al* combined Raman spectroscopy with optical contrast measurements and demonstrated statistically that bilayer and other FLG samples can also exhibit a Lorentzian 2D band, in some cases with a bandwidth even narrower than that of SLG [83]. Although this result addressed discrepancies already hinted at in earlier studies, it has received little attention in the literature.

Furthermore, the relative intensity of the 2D band with respect to the G band can also depend on the excitation wavelength, much like the D band, although this effect is not yet well documented in the literature. Since the D and 2D bands share certain matrix elements in their Raman cross-sections, the interpretation of spectra becomes even more complex [84].

3.3. Raman features of graphene stacks

When a twist angle is introduced between two graphene layers, it was already shown in 2008, only two years after the first Raman spectra of AB-stacked FLG, that the 2D band profile can become a single-mode Lorentzian shape, similar to monolayer graphene [47]. It demonstrated that twisting layers modifies the interlayer coupling. Depending on the twist angle, the layers can become electronically decoupled, resulting in band structures that are effectively indistinguishable from those of single-layer graphene (SLG) [85]. The same effect was later confirmed in two independent studies published in 2012 [13, 56].

In the case of tBLG, van Hove resonances modify strongly their Raman response, and the effect is highly sensitive to the excitation wavelength:

- **G-band intensity** can be enhanced by up to 2 orders of magnitude for specific combinations of laser wavelength and twist angle, which in some cases undermines the use of the I_D/I_G ratio as a reliable measure of defect density, or the I_{2D}/I_G ratio as a reliable measure of the doping [76].
- **2D-band intensity** can increase by a factor of 3 when moving from small twist angles ($0\text{--}10^\circ$) to larger angles ($20\text{--}30^\circ$) [13].
- **2D-band position** may shift upward by as much as 20 cm^{-1} relative to monolayer graphene, although for 633 nm excitation it remains nearly unchanged at $\sim 7\text{--}8^\circ$ [13].
- **2D-band linewidth** can reach values as high as 70 cm^{-1} , but for twist angles above 10° it decreases toward the typical Lorentzian value observed in monolayer graphene [13].

These observations have been confirmed in several subsequent studies [16, 86, 87], which reported I_{2D}/I_G ratios as high as 4–5, 2D bandwidths down to 17 cm^{-1} , and upshifts of the 2D band by about 8 cm^{-1} . If neither the I_{2D}/I_G ratio nor the 2D bandwidth alone can reliably indicate the number of graphene layers, an important question arises: is Raman spectroscopy still a quick and useful tool to assess the quality of graphene produced by a given process? The answer is likely yes, but it requires more attention in the analysis and a comprehensive evaluation of all spectroscopic parameters. We return to this point in the final section of this paper.

4. Effect of defects, twist, number of layers, doping and strain

In this section, before presenting a general methodology for determining the number of layers in graphene samples using Raman spectroscopy, we will proceed step by step to highlight the challenges involved. We then propose a structured strategy to reliably identify the number of layers.

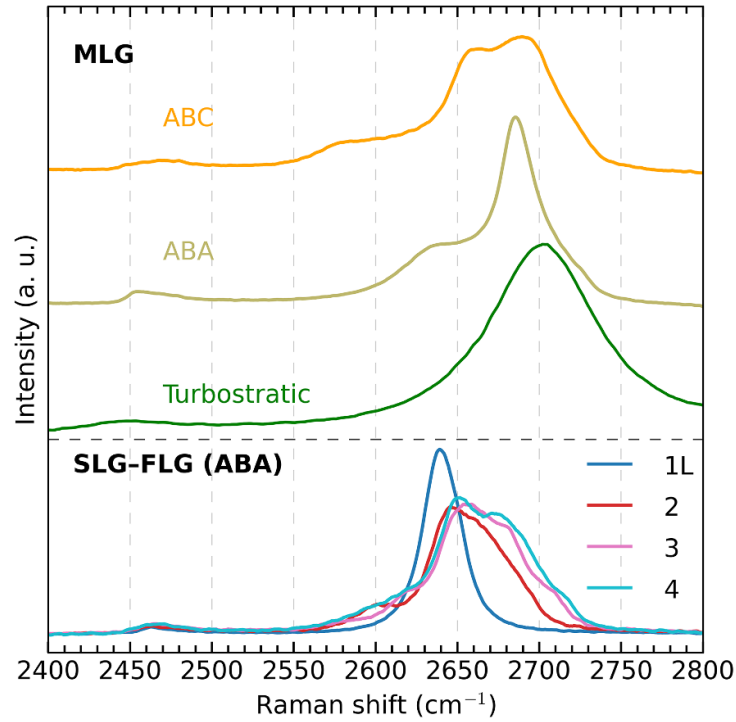


Figure 3. Raman 2D-band spectra of graphene recorded with a 633 nm excitation. The lower panel shows single-layer graphene (SLG) and few-layer graphene (FLG) with ABA stacking, measured on 90 nm SiO₂ under identical acquisition conditions, without normalization. The upper panel shows multilayer graphene (MLG) corresponding to different stacking orders: ABC, ABA and turbostratic. The MLG spectra are normalized and vertically shifted for clarity.

Figure 3 presents the Raman 2D band measured in MLG and in SLG-FLG. In figure 4, we present four typical Raman spectra and invite the reader to anticipate which spectrum corresponds to which sample. The spectra originate from different sources of graphene: Spectra 1 and 2 were obtained from two different regions of the same exfoliated graphene flake, as shown in the inset of figure 5. Sample 1 corresponds to a graphene layer folded onto itself, while Sample 2 originates from the flat graphene region. Spectrum 3, corresponds to a modified CVD-graphene by AFM cutting/folding method [76] and spectrum 4 was measured from a CVD-grown graphene sample. All four samples were transferred onto a Si/SiO₂ substrate measured in air, and Raman spectra were obtained using a Horiba Jobin Yvon HR800 setup with an excitation wavelength of $\lambda_L = 633$ nm, a $100\times$ objective (numerical aperture of 0.9), and a $600\text{ grooves mm}^{-1}$ grating. The spectral resolution was about 1 cm^{-1} . The laser power was kept below $1\text{ mW }\mu\text{m}^{-2}$ to prevent sample damage. For reference, the vertical dashed lines in the figure 4 indicate the positions of the G and 2D bands for undoped and unstrained SLG. Under 633 nm excitation, these bands are expected at 1582 cm^{-1} (G band) and 2630 cm^{-1} (2D band), while under 514 nm excitation they are expected at 1582 cm^{-1} (G band) and 2675 cm^{-1} (2D band). Additionally, several bandwidths were extracted by fitting the peaks with Lorentzian profiles.

Below, we provide a set of incorrect interpretations, followed by justifications that clarify why certain interpretations do not match the data:

- For samples 1–4, since their 2D bands can be fitted with Lorentzian functions, they must all correspond SLG.
- **Incorrect.** Turbostratic graphite (see figure 3) also exhibits a Lorentzian-shaped 2D band. Thus, the presence of a Lorentzian profile alone is insufficient to conclude that a sample is SLG.
- For sample 4, since $I_{2D}/I_G \approx 3$ and the 2D band exhibits a Lorentzian shape, it must be SLG.
- **Incorrect.** While a high ratio I_{2D}/I_G together with a Lorentzian 2D band is often sufficient to identify SLG, in this case the 2D bandwidth is significantly broader than expected. This broadening arises from a twist angle between stacked layers, indicating that this sample is not a true SLG.

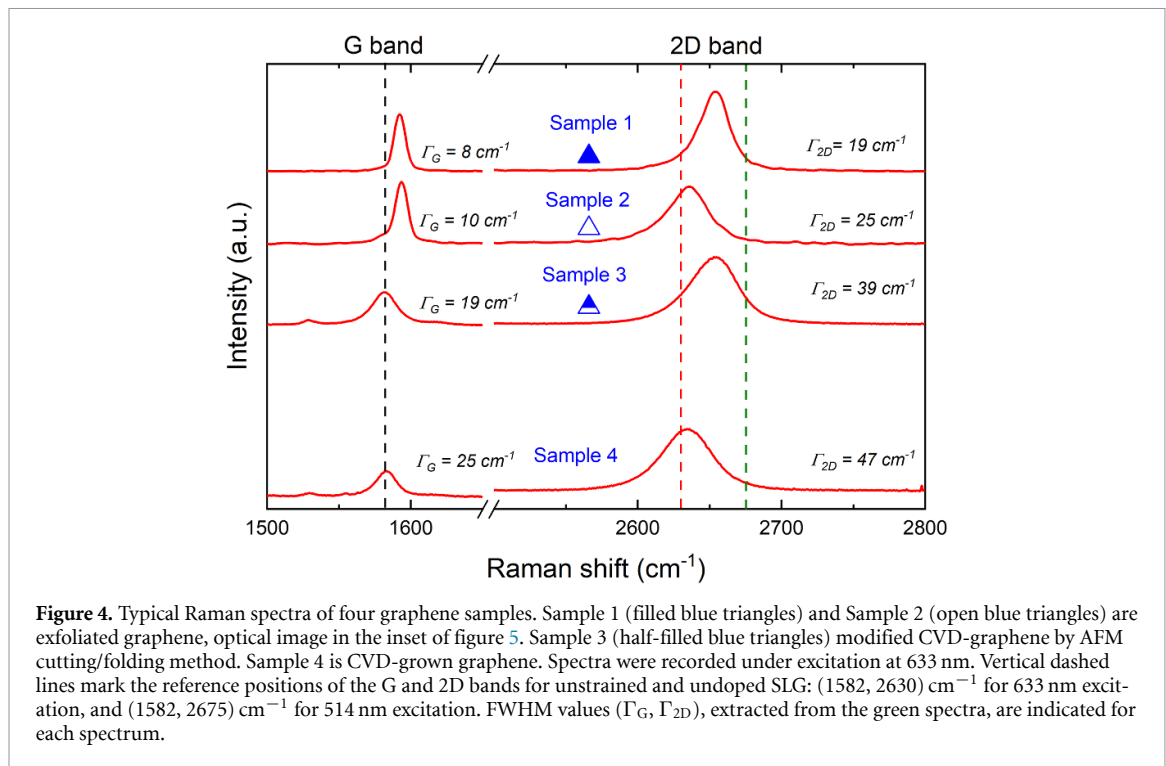


Figure 4. Typical Raman spectra of four graphene samples. Sample 1 (filled blue triangles) and Sample 2 (open blue triangles) are exfoliated graphene, optical image in the inset of figure 5. Sample 3 (half-filled blue triangles) modified CVD-graphene by AFM cutting/folding method. Sample 4 is CVD-grown graphene. Spectra were recorded under excitation at 633 nm. Vertical dashed lines mark the reference positions of the G and 2D bands for unstrained and undoped SLG: (1582, 2630) cm^{-1} for 633 nm excitation, and (1582, 2675) cm^{-1} for 514 nm excitation. FWHM values (Γ_G , Γ_{2D}), extracted from the green spectra, are indicated for each spectrum.

- For sample 1, the 2D bandwidth is 19 cm^{-1} , which is very close to that of SLG, and the band is Lorentzian, therefore, it must be a single layer.
- **Incorrect.** Although the 2D bandwidth is narrow, the low ratio $I_{2D}/I_G \approx 1.6$ indicates significant doping. The positions of the G and 2D bands, as well as the G-band width, further confirm this. Since doping also reduces the 2D bandwidth, the narrow line shape here is a consequence of doping rather than definitive evidence of SLG.
- For sample 2, the 2D bandwidth is relatively large and the I_{2D}/I_G ratio is less than 3, therefore, it cannot be an SLG.
- **Incorrect.** Counterintuitively, this spectrum is the only one in figure 4 that corresponds to an SLG. The discrepancy arises from doping, which alters the positions, widths, and relative intensity of the Raman bands. At this stage identifying the sample as SLG is not straightforward, and confirmation requires additional criteria, as shown in figures 5 and 6.
- For sample 3, because its spectrum looks like that of sample 2, it should be a doped SLG.
- **Incorrect.** The 2D band is strongly shifted while the G band remains essentially unshifted. This inconsistent with doping which would affect the G band position and width. Instead it points to a reduction of the Fermi velocity (v_F) caused by the interlayer coupling in few layers graphene stack [48]. The modified electronic dispersion changes the resonance condition, which gives a 2D shift band without a corresponding G shift. Thus, sample 3 is a FLG, not SLG.

Although the four spectra presented earlier are very similar, several methods exist to estimate the number of layers [88]. Some are geometry-dependent, such as light transmission [89], while others rely on the intensities of the G or 2D modes, which can be affected by resonance effects and thus lose reliability. Among these approaches, the attenuation of the underlying silicon band offers a simple and robust criterion to determine the layer number with confidence [90, 91]. In our case, with a 90 nm SiO_2 substrate, the layer number can be determined from the attenuation of the silicon band.

In figure 5, we display the Raman spectra of sample 1 (orange), sample 2 (blue), and the pristine substrate (black), all recorded under identical experimental conditions. The inset optical image locates the measurement spots: the orange spectrum is taken from a folded region of the same flake sampled at the blue spot, whereas the black spot lies on bare substrate (no G or 2D band). Because the sheet

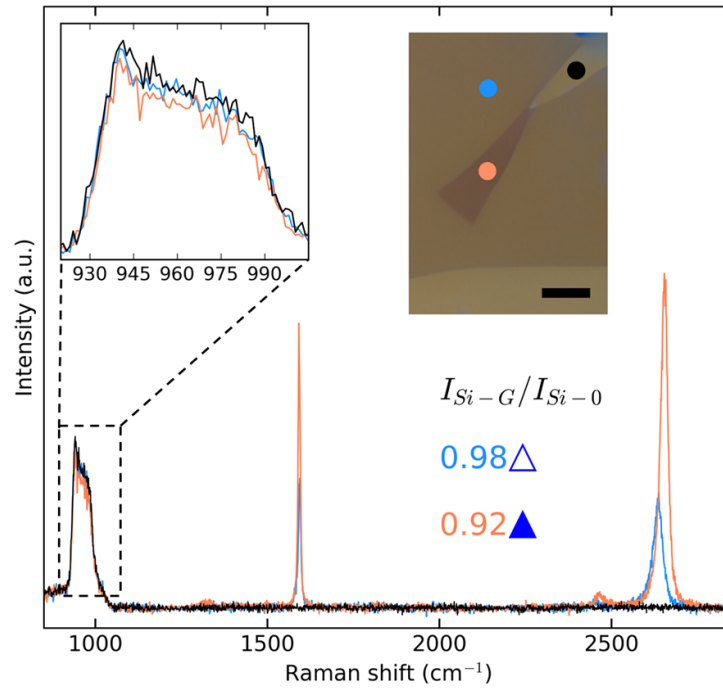
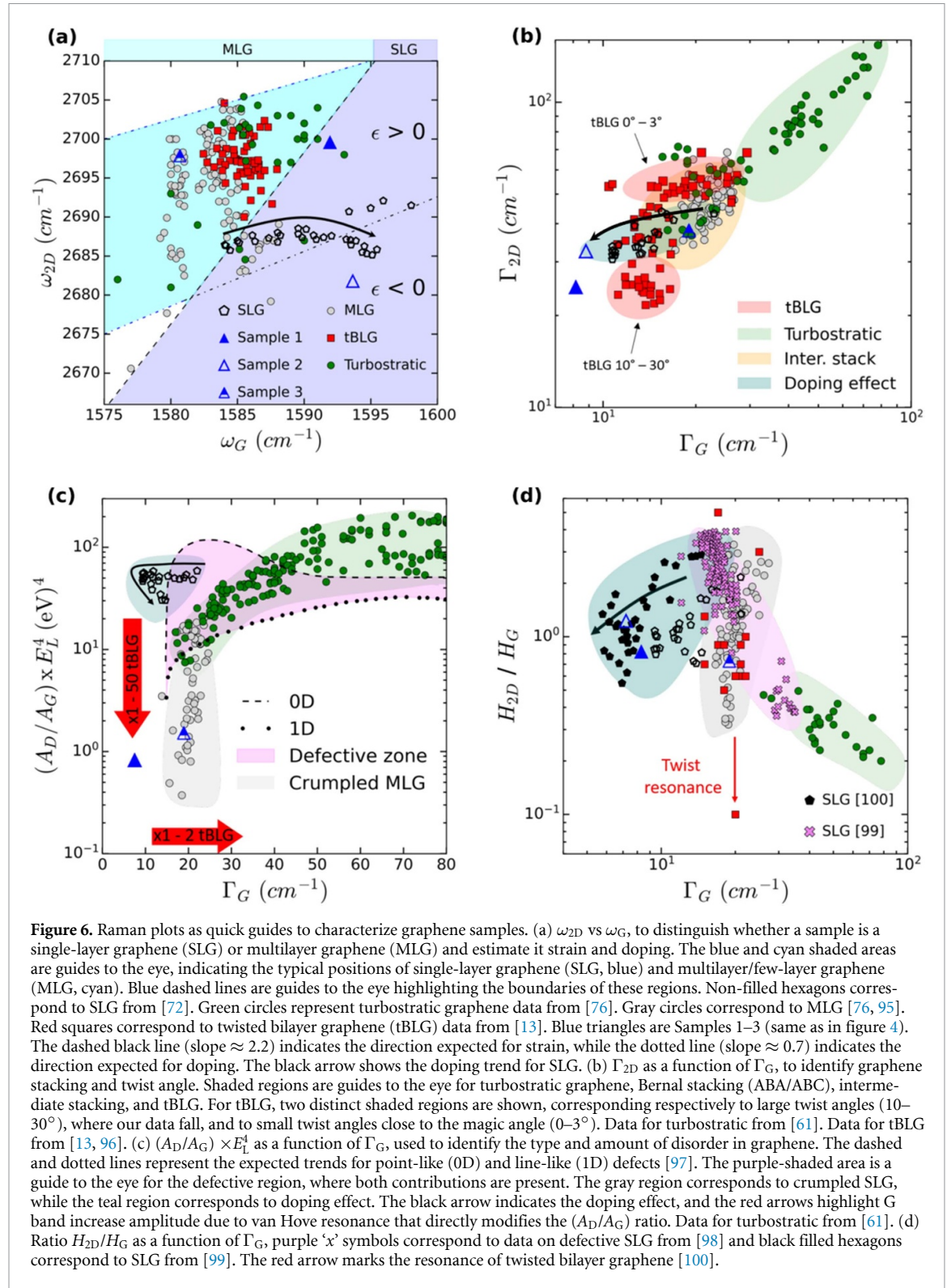


Figure 5. Attenuation of the underlying silicon signal by graphene. Raman spectra are shown for Sample 1 (orange) and Sample 2 (light blue). The black spectrum corresponds to the reference silicon signal. The colored dots in the optical image in the inset mark the exact position where the Raman spectra were measured. The attenuation is quantified by the ratio $I_{\text{Si-G}}/I_{\text{Si-0}}$ giving 0.92 for sample 1 and 0.98 for sample 2. The inset zoom on the silicon peak highlights the difference in intensity between the two regions. The black scale bar in the optical image is $4\ \mu\text{m}$. Raman map analysis is presented in figure SI 3 in supplementary information.

folds back onto itself, the number of layers is higher than the adjacent region. From these three spectra, the silicon band attenuation factors are 0.98 and 0.92 for samples 1 and 2, respectively. Using the calibration of Li *et al* [90, 92], we therefore confirm that sample 2 is a SLG and sample 1 is a BLG. Similar approaches based on substrate peak attenuation have also been used for different sources of graphene. No *et al* [91] applied Si peak analysis for CVD-grown MLG on Si/SiO₂ and Dobrowolski *et al* demonstrated a similar attenuation approach using the Raman response of the SiC substrate for epitaxial graphene [93]. For sample 1, the 2D band is well described by a single Lorentzian rather than the four-component profile expected for AB stacked bilayer graphene [11], indicating it is not Bernal-stacked. $I_{2\text{D}}/I_{\text{G}}$ ratio can, in principle, serve as a complementary indicator, but it is strongly affected by resonance. One question that may arise is whether the regions identified as SLG and BLG are truly homogeneous. To address this concern, we performed Raman mapping to examine the uniformity of each region. In the Supplementary Information, we present detailed maps acquired at 633 nm excitation for the sample shown in the inset, using several parameters, such as the normalized G-band area, the G- and 2D-band linewidths, the D-to-G band area ratio, the D-band area, and the 2D-to-G band area ratio (see figure SI 3 in the supplementary information). We also provide additional data from another sample based on the peak-height ratio between the G band and the Si substrate peak (see figure SI 2 in supplementary information). Other optical techniques can be coupled with Raman spectroscopy to strengthen the conclusions [94]. For more complex cases and when only Raman is available it may be necessary to account for multiple effect (presence of defects, twist angle, number of layers, doping and strain) that modify the graphene and, consequently, the spectroscopic parameters.

To make this practical, figure 6 shows four concise plots to determine the layer number and, in particular, for testing whether a Lorentzian 2D band truly confirms SLG. Although these effects (number of layers, defects, twist angle, doping and strain) are all Raman observables. Taken together, these plots make the interpretation clearer and more reliable. Here we consider the G- and 2D-band positions (ω_{G} , $\omega_{2\text{D}}$), their bandwidths (Γ_{G} , $\Gamma_{2\text{D}}$), the peak-height ratio $H_{2\text{D}}/H_{\text{G}}$ (we use ‘*H*’ for height, note that ‘*T*’ is sometimes used for height or for integrated area) [68], and the area ratio $A_{\text{D}}/A_{\text{G}}$. We focus on parameters that are well documented, we do not consider polarization effects here, although they can be helpful [92]. The four pairs of parameters presented show how the listed physical effects influence these features. All spectra in this dataset exhibit a Lorentzian 2D band. Broadening by the apparatus is possible when using low-dispersion gratings. However since the intrinsic widths of the 2D and G bands are generally



larger than 10 cm^{-1} , this effect can be neglected. Most of the spectra shown in this figure were acquired using gratings with at least $600 \text{ lines mm}^{-1}$. Most of the data points in this figure were recorded with 514 nm excitation. Because the spectra presented in figure 4 were obtained at 633 nm , we applied a correction of 45 cm^{-1} to align them with the 514 nm data as 2D band is dispersive ($100 \text{ cm}^{-1} \text{ eV}^{-1}$). The spectra exhibit a Lorentzian profile, which we also confirm for the measurements at 514 nm and 532 nm (see supplementary information).

Figure 6(a) uses (ω_G, ω_{2D}) map to estimate strain and doping at the same time and to determine whether a sample is SLG. Points that follow the line of slope ≈ 0.7 indicate a doping dominated behavior, while those along the slope ≈ 2.2 indicate a strain-dominated behavior [25, 26, 101]. For SLG,

combination of strain and doping places the data on the blue side (right) of the diagonal. The points on the cyan side (left) of the diagonal are not SLG and instead indicate multilayer/turbostratic systems. tBLG [13, 102] and MLG formed by AFM cutting/folding method [76, 95] fall in this cyan region. In our data, sample 3 lies in this MLG region, sample 2 falls in the SLG, and sample 1 is near the boundary between the two regions. This boundary is a practical guide rather than a strict SLG/FLG cutoff.

Figure 6(b) plot Γ_{2D} versus Γ_G . These widths tend to be positively correlated (an empirical trend), but specific perturbations induce systematic deviations. In this figure, all spectra that are close to Lorentzian shapes are fitted with Lorentzian functions. Regarding tBLG, data from [102] cannot be included, as they were fitted with one to three components. Large twist angles (with Γ_{2D} in the range of $20\text{--}30\text{ cm}^{-1}$ according to [13]) are located in a distinct region compared to small twist angles [96] (close to the magic angle, i.e. $0^\circ\text{--}3^\circ$). Data corresponding to [96] have been estimated using a single Lorentzian, although multicomponent features are observed near the magic angle, which we do not discuss here. Moderate doping primarily reduces Γ_G and, to a lesser extent, narrows Γ_{2D} [99]. In tBLG, the G-band linewidth (Γ_G) changes with doping depends on the twist angle: near the magic angle, n and p doping affect Γ_G asymmetrically, whereas at larger angles this asymmetry vanishes [103]. To place the data of [99] on this plot, we used the fact that both G and 2D bands are Lorentzian and that H_{2D}/H_G , A_G/A_{2D} , and Γ_G were reported; the relation $H = 2A/\pi\Gamma$ then allowed us to determine Γ_{2D} . We label an intermediate region populated by tBLG and by AFM cutting/folding multilayers, distinct from the turbostratic region. There is an overlap between turbostratic, this intermediate region, and doped SLG. The shaded areas are guides to the eye, highlighting where data points are most densely concentrated. In our data, sample 2 lies at the edge of the doped-SLG zone (consistent with SLG), for sample 1 this plot alone cannot indicate the layer number, though it indicates doping and sample 3 locates within the intermediate region. Note that for tBLG, the shape of figure 6(b) could change slightly depending on the laser used, as resonance effects mainly influence the dependence of Γ_{2D} on the twist angle [13], and, to a lesser extent, Γ_G [104].

Figure 6(c) shows the ratio $A_D/A_G \times E_L^4$, where E_L is the laser excitation energy. The purple zone marks defective SLG [97] obtained by combination of point (0D) and grain boundary (1D) defects. This purple zone overlaps strongly with turbostratic samples however the A_D/A_G ratio alone cannot separate them. In defective SLG, moderate doping mainly lowers Γ_G (leftward shift) and higher doping result in vertical downshift of A_D/A_G [99]. Because systematic data on how A_D/A_G varies with doping at fixed defect density are limited in the literature, the doping region drawn here is provisional and should be better constrained in future work. Crumpled graphene including AFM cutting/folding multilayers shows very low A_D/A_G due to resonance-enhanced G-band intensity, similar to tBLG [76], twisting typically reduces A_D/A_G by $\sim 1\text{--}2$ orders of magnitude and increases Γ_G by about a factor of $1\text{--}2$. In this plot, twist and doping act partly in opposite directions, which helps separate their contributions. For sample 1, the silicon attenuation confirms that it is a bilayer. In figure 6(a), it falls within the boundary region between the SLG and MLG zones. Figure 6(b) shows clear evidence of doping, which is further supported by figure 6(c). The strong vertical downshift observed in figure 6(c) indicates an enhancement of the G band, consistent with resonance effects typically found in twisted graphene layers. This suggests that sample 1 is not a conventional bilayer but rather a tBLG. Sample 3 lies in the crumpled region and sample 2 is not shown because its defect density is likely very low ($A_D/A_G \approx 0$), making it invisible on the logarithmic scale.

Figure 6(d) plots the peak-height ratio H_{2D}/H_G versus the G-band width Γ_G . As defects increase in SLG, points move diagonally toward the turbostratic domain [97]. The tBLG [100] and AFM cutting/folded multilayers, which shift mainly vertically because twist-induced resonance strongly enhances the G band the 2D-band intensity changes only within a factor of two, giving a very low H_{2D}/H_G even at low Γ_G . Doping moves both non-defective [72] and defective SLG [99] into a broad left region (smaller Γ_G). In our data, samples 1 and 2 fall in this doping-influenced region.

The approach above give reliable layer determination and capture the main effects. In occasional ambiguous cases when several effects overlap additional criteria may be examined such as the LO+LA and LO+TA combination modes. These features are about two orders of magnitude weaker than the main bands, so we do not analyze them here [105–107].

5. Conclusion

Raman spectroscopy has remained the primary tool for characterizing graphene-based samples. Decades of work have progressively refined its interpretation by accounting for many subtle effects. Yet a persistent source of error is the misinterpretation of the number of layers. As we have shown, this mistake arises largely because twist effects between layers are often overlooked. We have reviewed the historical

development of the literature. In practice, it is often challenging to identify works that explicitly address stacking faults, whether in early or more recent studies. In this article, we emphasize that the 2D band profile can appear Lorentzian even in samples that are not SLG. Since Raman spectra carry information about of interlayer interactions, Raman spectroscopy still has strong potential as a future tool, particularly in view of the following research:

- The twist angle of heterostructures containing graphene in microdevices can now be precisely controlled [108, 109].
- Graphite has come back into focus, precisely because of stacking-related phenomena [110].
- Nano-Raman starts to be used [111].
- Superconductivity is observed in graphene with increasingly stacking configurations [112].

Some automated methods used to identify the number of layers in mass-produced flakes [88] may fail, since they often rely on the assumption that only the monolayer exhibits a Lorentzian 2D band, and because defects can also broaden this feature [113]. Artificial intelligence (AI) offers a powerful alternative approach [114], as machine-learning models can be trained to identify the characteristic Raman and optical features associated with different twist angles and stacking configurations. By learning the relationships between spectral shapes, peak positions, band widths, and their evolution with twist angle, these models can systematically distinguish structural configurations. Recent studies have demonstrated that trained machine learning algorithms can classify twist-angle ranges, identify stacking order [115–119].

This work illustrates the difficulty of locating specific information in a field characterized by a vast number of publications and complex terminology. Similar challenges are likely to be encountered in other research areas as well. The use of generative AI for bibliographic research could offer valuable support to researchers, and recent tests in other domains have already shown promising results [120]. We believe that our work will help researchers in more easily characterizing their samples. For instance, the role of the 2D bandwidth in evaluating crystal quality in hBN-encapsulated graphene has been highlighted, as it correlates directly with carrier mobility [121].

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

Supplementary information available at <https://doi.org/10.1088/1361-6633/ac3df8/data1>.

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